

Today •

Today

- Model reduction in a nutshell
- Formulation of full model

Announcements

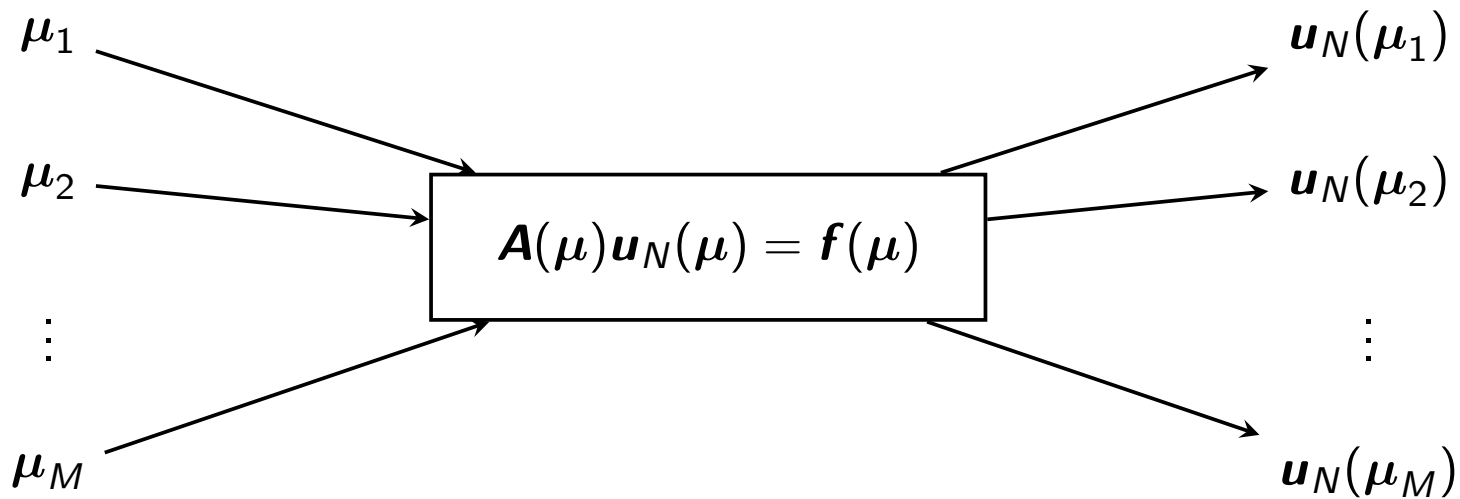
- Homework 1 will be posted today; due 16.10

Recap: Abstract full model

High-fidelity parametric model

$$\mathbf{A}(\boldsymbol{\mu})\mathbf{u}_N(\boldsymbol{\mu}) = \mathbf{F}(\boldsymbol{\mu}), \quad \mathbf{u}_N(\boldsymbol{\mu}) \in \mathbb{R}^N, \quad \boldsymbol{\mu} \in \mathcal{D} \subset \mathbb{R}^p$$

- State of system: $\mathbf{u}_N(\boldsymbol{\mu})$
- Dimension of state: $N \in \mathbb{N}$
- Parameter of system: $\boldsymbol{\mu} \in \mathcal{D}$
- System matrix $\mathbf{A}(\boldsymbol{\mu}) \in \mathbb{R}^{N \times N}$; invertible



Recap: Solution manifold

Considering all states that can be generated by varying the parameter leads to the solution manifold

$$\mathcal{M}_N = \{\mathbf{u}_N(\boldsymbol{\mu}) \mid \boldsymbol{\mu} \in \mathcal{D}\} \subset \mathbb{R}^N.$$

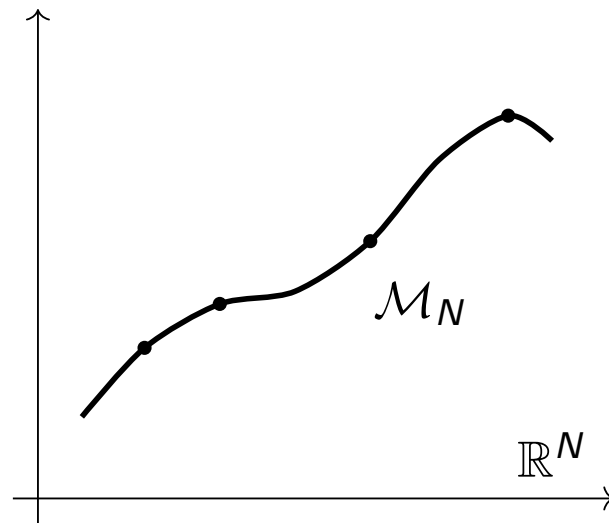
- Contains all solutions of

$$\mathbf{A}(\boldsymbol{\mu})\mathbf{u}_N(\boldsymbol{\mu}) = \mathbf{F}(\boldsymbol{\mu}), \quad \boldsymbol{\mu} \in \mathcal{D}.$$

- Ambient dimension can be very large

$$N \gg 1.$$

Model reduction: Exploit structure of $\mathcal{M}_N$.



Recap: Low-dimensional approximation of the solution manifold

We seek a subspace

$$\mathcal{V}_n \subset \mathbb{R}^N, \quad \dim(\mathcal{V}_n) = n, \quad n \ll N,$$

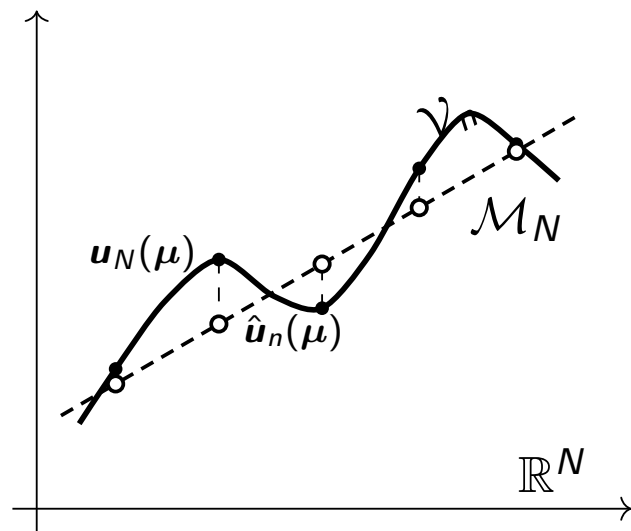
that approximates the solution manifold. Then, for every parameter $\mu \in \mathcal{D}$, seek an approximation

$$\hat{\mathbf{u}}_n(\mu) \in \mathcal{V}_n$$

of the high-dimensional solution $\mathbf{u}_N(\mu) \in \mathbb{R}^N$.

- Space $\mathcal{V}_n$ is fixed and does not depend on μ
- Different parameters lead to different elements

$$\hat{\mathbf{u}}_n(\mu_1), \dots, \hat{\mathbf{u}}_n(\mu_M) \in \mathcal{V}_n$$



Curved solution manifold $\mathcal{M}_N$ and linear reduced space $\mathcal{V}_n$.

Recap: From the ideal space to a computable space

A Kolmogorov-optimal space is generally not available computationally.

Instead, replace the complete solution manifold by a finite set of representative full-model solutions

$$\mathcal{M}_N = \{\dots\} \quad \longrightarrow \quad \mathbf{u}_N(\boldsymbol{\mu}_1), \dots, \mathbf{u}_N(\boldsymbol{\mu}_M).$$

for M training parameters $\boldsymbol{\mu}_1, \dots, \boldsymbol{\mu}_M \in \mathcal{D}$

This set of **snapshots** provides information about the solution manifold.

We can now ask a computable version of the approximation problem,

$$\min_{\substack{\mathcal{V}_n \subset \mathbb{R}^N \\ \dim(\mathcal{V}_n) = n}} \frac{1}{M} \sum_{i=1}^M \min_{\mathbf{v} \in \mathcal{V}_n} \|\mathbf{u}_N(\boldsymbol{\mu}_i) - \mathbf{v}\|_2^2$$

Find the n -dimensional space that best represents the training data.

Recap: Snapshots

For training parameters

$$\mathcal{D}_{\text{train}} = \{\boldsymbol{\mu}_1, \dots, \boldsymbol{\mu}_M\} \subset \mathcal{D},$$

compute the full-model states

$$\mathbf{u}_N(\boldsymbol{\mu}_1), \dots, \mathbf{u}_N(\boldsymbol{\mu}_M).$$

Collect them in the **snapshot matrix**

$$\mathbf{Y} = \left[\begin{array}{c|c|c|c} & & & \\ \mathbf{u}_N(\boldsymbol{\mu}_1) & \mathbf{u}_N(\boldsymbol{\mu}_2) & \cdots & \mathbf{u}_N(\boldsymbol{\mu}_M) \\ & & & \end{array} \right] \in \mathbb{R}^{N \times M}.$$

Key question: How should the training parameters be chosen?

Recap: Proper orthogonal decomposition (POD)

Solution to approximation problem

$$\min_{\substack{\mathcal{V}_n \subset \mathbb{R}^N \\ \dim(\mathcal{V}_n) = n}} \frac{1}{M} \sum_{i=1}^M \min_{\mathbf{v} \in \mathcal{V}_n} \|\mathbf{u}_N(\boldsymbol{\mu}_i) - \mathbf{v}\|_2^2$$

given by the the singular value decomposition

$$\mathbf{Y} = \mathbf{V}\boldsymbol{\Sigma}\mathbf{W}^T, \quad \sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0,$$

where

$$\mathbf{V} = [\mathbf{v}_1, \dots, \mathbf{v}_r], \quad r = \text{rank}(\mathbf{Y}).$$

Then the optimal n -dimensional space is

$$\mathcal{V}_n^{\text{POD}} = \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_n\}.$$

The corresponding minimum error is

$$\frac{1}{M} \sum_{i=1}^M \|\mathbf{u}_N(\boldsymbol{\mu}_i) - \mathcal{P}_n \mathbf{u}_N(\boldsymbol{\mu}_i)\|_2^2 = \frac{1}{M} \sum_{j=n+1}^r \sigma_j^2.$$

Recap: From a basis to a reduced model

POD gives us a reduced space $\mathcal{V}_n = \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$

For a new parameter $\boldsymbol{\mu} \in \mathcal{D}$, we now seek a solution of the linear system in $\mathcal{V}_n$.

Every element of $\mathcal{V}_n$ can be represented via a reduced state $\mathbf{u}_n(\boldsymbol{\mu}) \in \mathbb{R}^n$,

$$\mathbf{V}\mathbf{u}_n(\boldsymbol{\mu}) \in \mathcal{V}_n$$

Insert this representation into the full-model equations

$$\mathbf{A}(\boldsymbol{\mu})\mathbf{u}_N(\boldsymbol{\mu}) = \mathbf{F}(\boldsymbol{\mu})$$

gives the residual

$$\mathbf{r}_N(\mathbf{u}_n; \boldsymbol{\mu}) = \mathbf{F}(\boldsymbol{\mu}) - \mathbf{A}(\boldsymbol{\mu})\mathbf{V}\mathbf{u}_n(\boldsymbol{\mu}) \in \mathbb{R}^N.$$

We have only n unknowns, but N equations. How should $\mathbf{u}_n(\boldsymbol{\mu})$ be chosen?

Recap: Galerkin projection

Galerkin condition requires the residual to be orthogonal to the reduced space

$$\mathbf{r}_N(\mathbf{u}_n; \mu) \perp \mathcal{V}_n.$$

Since

$$\mathcal{V}_n = \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_n\},$$

orthogonality condition can be written as

$$\mathbf{v}_j^T \mathbf{r}_N(\mathbf{u}_n; \mu) = 0, \quad j = 1, \dots, n,$$

or, in matrix form, $\mathbf{V}^T \mathbf{r}_N(\mathbf{u}_n; \mu) = 0$.

Insert the residual

$$\mathbf{V}^T [\mathbf{F}(\mu) - \mathbf{A}(\mu) \mathbf{V} \mathbf{u}_n(\mu)] = 0.$$

Hence

$$\mathbf{V}^T \mathbf{F}(\mu) - \mathbf{V}^T \mathbf{A}(\mu) \mathbf{V} \mathbf{u}_n(\mu) = 0,$$

and therefore

$$\underbrace{\mathbf{V}^T \mathbf{A}(\mu) \mathbf{V}}_{\mathbf{A}_n(\mu) \in \mathbb{R}^{n \times n}} \mathbf{u}_n(\mu) = \underbrace{\mathbf{V}^T \mathbf{F}(\mu)}_{\mathbf{F}_n(\mu) \in \mathbb{R}^n}.$$

Recap: Galerkin as a best approximation I

Assume that $\mathbf{A}(\mu)$ is symmetric positive definite and define the error

$$\mathbf{e}(\mu) = \mathbf{u}_N(\mu) - \mathbf{V}\mathbf{u}_n(\mu).$$

Since

$$\mathbf{A}(\mu)\mathbf{u}_N(\mu) = \mathbf{F}(\mu),$$

the residual can be written as

$$\begin{aligned}\mathbf{r}_N(\mathbf{u}_n; \mu) &= \mathbf{F}(\mu) - \mathbf{A}(\mu)\mathbf{V}\mathbf{u}_n(\mu) \\ &= \mathbf{A}(\mu)\mathbf{u}_N(\mu) - \mathbf{A}(\mu)\mathbf{V}\mathbf{u}_n(\mu) \\ &= \mathbf{A}(\mu)\mathbf{e}(\mu).\end{aligned}$$

The Galerkin condition $\mathbf{V}^T \mathbf{r}_N(\mathbf{u}_n; \mu) = 0$ therefore gives $\mathbf{V}^T \mathbf{A}(\mu)\mathbf{e}(\mu) = 0$ and thus

$$\mathbf{v}^T \mathbf{A}(\mu)\mathbf{e}(\mu) = 0 \quad \forall \mathbf{v} \in \mathcal{V}_n.$$

Because $\mathbf{A}(\mu)$ is symmetric positive definite, we can define the inner product

$$\langle \mathbf{x}, \mathbf{y} \rangle_{\mathbf{A}(\mu)} = \mathbf{x}^T \mathbf{A}(\mu) \mathbf{y}$$

and obtain

$$\mathbf{e}(\mu) \perp_{\mathbf{A}(\mu)} \mathcal{V}_n.$$

The Galerkin solution selects $\mathbf{u}_n$ so that the error is $\mathbf{A}(\mu)$ -orthogonal to $\mathcal{V}_n$.

Galerkin gives best-approximation in this case

↳ consider

$$\min_{z \in \mathbb{R}^n} \frac{1}{2} \| \underbrace{u_N(\mu) - V_h z}_{= e(\mu)} \|_{A(\mu)}^2$$

↳ Objective: $J(z) = \frac{1}{2} (u_N(\mu) - V_h z)^T A(u_N(\mu) - V_h z)$

Expand (use $A=A^T$):

$$J(z) = \frac{1}{2} u_N^T A u_N - u_N^T A V_h z + \frac{1}{2} z^T V_h^T A V_h z$$

Gradient

$$\nabla_z J(z) = -V_h^T A u_N + V_h^T A V_h z$$

$$\text{Set } \nabla_z J(z) = 0$$

$$V_h^T A V_h z = V_h^T \underbrace{A u_N}_F = V_h^T F$$

→ Galerkin system

→ $V_h^T A V_h$ is spd $\leadsto$ unique solution
 $\leadsto u_h$

Residual minimization:

$$e = u_N - V_h u_h, \quad r = F - A V_h u_h = A e$$

$$\begin{aligned} \|e\|_A^2 &= e^T A e = e^T A^T A^{-1} A e = (A e)^T A^{-1} (A e) \\ &= r^T A^{-1} r = \|r\|_{A^{-1}}^2 \end{aligned}$$

Galerkin is

→ best approximation in A -norm

→ minimal residual approximation in A' -norm

Non-symmetric system

↳ A does not induce an inner product

↳ Galerkin fails

$$A = \begin{bmatrix} 0 & 1 \\ -2 & 1 \end{bmatrix}, \quad F = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

→ not sym.
but inv.

$$A u_N = F \quad \text{gives} \quad u_N = \begin{bmatrix} \frac{1}{2} \\ 1 \end{bmatrix}$$

↳ Now consider subspace

$$V_n = \text{Span} \left\{ \underline{\begin{bmatrix} 1 \\ 0 \end{bmatrix}} \right\}$$

$$\begin{aligned} r_N(u_n) &= F - A V_n u_n = \begin{bmatrix} 1 \\ 0 \end{bmatrix} - \begin{bmatrix} 0 & 1 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} u_n[1] \\ 0 \end{bmatrix} \\ &= \begin{bmatrix} 1 \\ 0 \end{bmatrix} - \begin{bmatrix} 0 \\ -2u_n[1] \end{bmatrix} \\ &= \begin{bmatrix} 1 \\ 2u_n[1] \end{bmatrix} \end{aligned}$$

Galerkin condition $V_n^T r_N(u_n) \stackrel{!}{=} 0$

$$\begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 2u_n[1] \end{bmatrix} = 1 \neq 0 \leadsto \text{no solution}$$

Residual minimization for var-syn systems

$$u_h = \underset{z \in \mathbb{R}^n}{\operatorname{argmin}} \frac{1}{2} \| \underbrace{F - AV_h z}_{r_N(u_h)} \|_2^2$$

Normal equations

$$(AV_h)^T (AV_h) z = (AV_h)^T F$$

$$\underbrace{V_h^T A^T AV_h}_{W} z = V_h^T A F$$

Write as residual

$$\underbrace{V_h^T A^T}_{W} (F - AV_h z) = 0$$

$$\underline{W^T AV_h z = W^T F}$$

Petrov-Galerkin condition

We obtain a different test space given by $\operatorname{Colspan}(W(\mu))$ important: $W(\mu)$ can depend on μ .

Now let us look at the error

$$\begin{aligned} \|r\|_2^2 &= \|F - AV_h z\|_2^2 = \|AV_N - AV_h z\|_2^2 \\ &= \|A(u_N - V_h z)\|_2^2 \\ &= (u_N - V_h z)^T A^T A (u_N - V_h z) \\ &= \|u_N - V_h z\|_{A^T A}^2 = \|e\|_{A^T A}^2 \end{aligned}$$

Best approximation in $\|\cdot\|_{A^T A}$

Solving the reduced system:

$$\underbrace{V^T A(\mu) V}_{\underline{A}_n(\mu) \in \mathbb{R}^{n \times n}} \underline{u}_n(\mu) = \underbrace{V^T F(\mu)}_{\underline{F}_n(\mu) \in \mathbb{R}^n}$$

↳ Because $n \ll N$, solving reduced system should be cheaper.

↳ Online phase (test phase, at inference)

① Assemble $A(\mu)$, $F(\mu)$

② Project

$$\underline{A}_n(\mu) = V^T A(\mu) V, \quad \underline{F}_n(\mu) = V^T F(\mu)$$

③ solve $n \times n$ system

↳ steps ① and ② involve full-model quantities
→ comp. expensive

↳ is solving $n \times n$ system always "cheap"?

→ Sparse vs dense

→ parallel

→ ...

small n does not imply faster
online computations than full model

Affine parameter dependence helps with ① and ②:

$$A(\mu) = \sum_{q=1}^{Q_A} \Theta_q^A(\mu) A_q, \quad F(\mu) = \sum_{q=1}^{Q_F} \Theta_q^F(\mu) F_q$$

with $A_1, \dots, A_{Q_A} \in \mathbb{R}^{N \times N}$, $F_1, \dots, F_{Q_F} \in \mathbb{R}^N$
 and Θ_q^A, Θ_q^F cheap to evaluate (indep. of N)
 and $Q_A, Q_F \ll N$.

Then

$$A_n(\mu) = \sum_{q=1}^{Q_A} \Theta_q^A(\mu) \underbrace{V^T A_q V}_{A_{n,q} \in \mathbb{R}^{n \times n}}$$

$$F_n(\mu) = \sum_{q=1}^{Q_F} \Theta_q^F(\mu) \underbrace{V^T F_q}_{F_{n,q} \in \mathbb{R}^n}$$

Offline (training): Compute and store

$A_{n,1}, \dots$

F_{n,Q_F}

Online : • assemble with costs indep. of N

$$A_n(\mu) u_n(\mu) = F_n(\mu)$$

$$\mathcal{O}(Q_A n^2)$$

$$\mathcal{O}(Q_F n)$$

• Solve with $\mathcal{O}(n^3)$

$$\text{total online: } \mathcal{O}(Q_A n^2 + Q_F n + n^3)$$

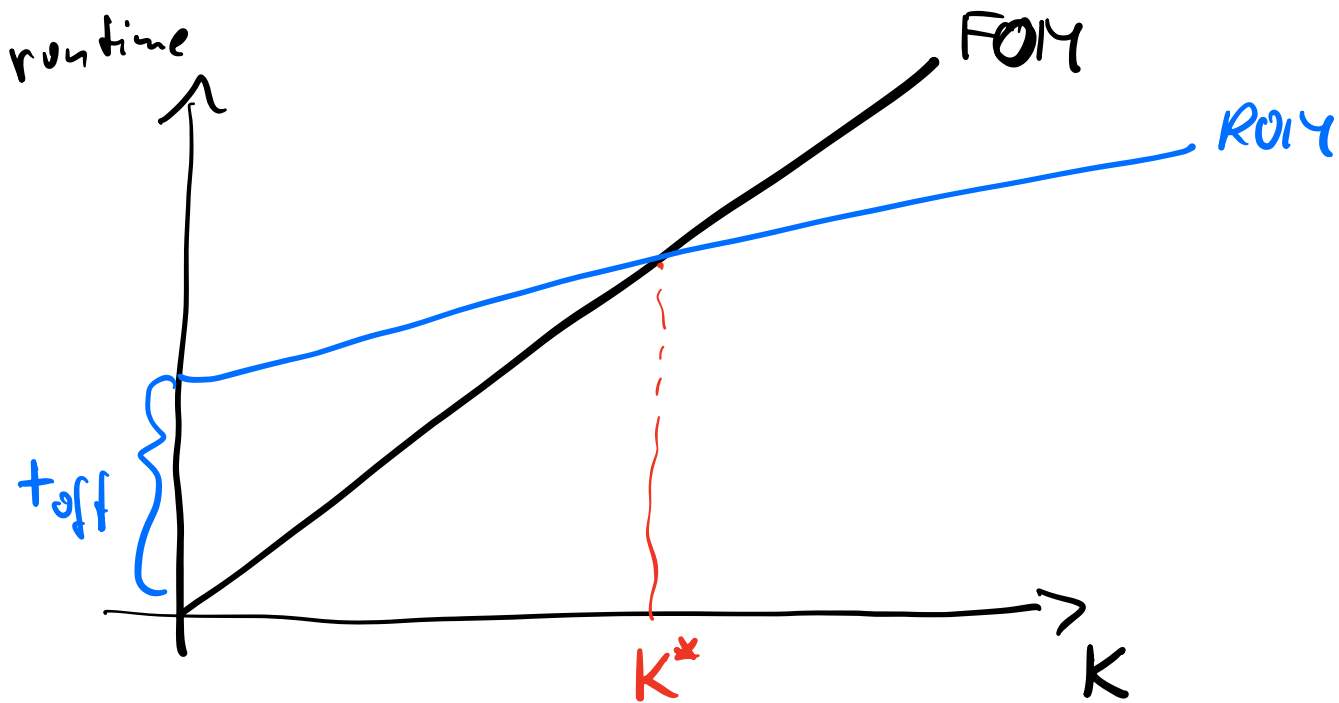
When does model reduction pay off?

↳ t_{FOM} ... costs of full model solve

↳ t_{off} ... construction of reduced model

↳ t_{on} ... costs of online solve with reduced model

↳ K ... number of parameters we want a solution of



Plan for the rest of today

1. Low-dimensional structure
2. Reduced-space construction
3. Projection-based reduced models
4. Online efficiency
5. Challenges and course outlook

Challenges

Intro: Kolmogorov barrier

Our reduced models use a fixed linear space

$$\mathcal{V}_n \subset \mathbb{R}^N, \quad \dim(\mathcal{V}_n) = n.$$

The best worst-case error achievable by *any* such space is the Kolmogorov n -width

$$d_n(\mathcal{M}_N) = \inf_{\substack{\mathcal{V}_n \subset \mathbb{R}^N \\ \dim(\mathcal{V}_n) = n}} \sup_{\mu \in \mathcal{D}} \inf_{\mathbf{v} \in \mathcal{V}_n} \|\mathbf{u}_N(\mu) - \mathbf{v}\|.$$

For favorable diffusion problems, the Kolmogorov n -width can decay exponentially fast,

$$d_n(\mathcal{M}_N) \leq Ce^{-cn}.$$

For transport- and wave-dominated problems with moving coherent structures, the decay can instead be only algebraic, for example

$$d_n(\mathcal{M}_N) \geq \frac{c}{\sqrt{n}}.$$

The decay of the Kolmogorov n -width determines the best accuracy possible with a fixed n -dimensional linear space.

Kolmogorov barrier: Decay is slow.

Intro: Bounding error of reduced solution

For a new parameter $\mu \in \mathcal{D}$, the reduced model gives $\mathbf{u}_n(\mu) \in \mathbb{R}^n$ and the corresponding approximation $\mathbf{V}\mathbf{u}_n(\mu) \in \mathbb{R}^N$.

What we would like to know is the error

$$\|\mathbf{u}_N(\mu) - \mathbf{V}\mathbf{u}_n(\mu)\|.$$

But evaluating this quantity requires the full-order solution

$$\mathbf{u}_N(\mu),$$

which is exactly what we wanted to avoid computing.

Thus, for a useful reduced model, we would like a quantity

$$\Delta_n(\mu)$$

that can be evaluated cheaply from reduced-model information and satisfies

$$\|\mathbf{u}_N(\mu) - \mathbf{V}\mathbf{u}_n(\mu)\| \leq \Delta_n(\mu).$$

Such an estimate should be

- computable without solving the full-order model,
- cheap to evaluate online (independently of N),
- informative for parameters μ that were not used for training.

Intro: Affine decomposition is unavailable

The offline/online decomposition relied on

$$\mathbf{A}(\boldsymbol{\mu}) = \sum_{q=1}^{Q_A} \theta_q^A(\boldsymbol{\mu}) \mathbf{A}_q$$

with small Q_A .

Such a representation may not exist with a sufficiently small number of terms, then, for a new $\boldsymbol{\mu}$,

$$\mathbf{A}_n(\boldsymbol{\mu}) = \mathbf{V}^T \mathbf{A}(\boldsymbol{\mu}) \mathbf{V}$$

may again require assembling and evaluating the full $N \times N$ operator.

A similar problem occurs for nonlinear models. Consider

$$\mathbf{A}(\boldsymbol{\mu}) \mathbf{u}_N(\boldsymbol{\mu}) + \mathbf{g}(\mathbf{u}_N(\boldsymbol{\mu}); \boldsymbol{\mu}) = \mathbf{F}(\boldsymbol{\mu}).$$

Projection gives the nonlinear reduced term

$$\boxed{\mathbf{V}^T \mathbf{g}(\mathbf{V} \mathbf{u}_n(\boldsymbol{\mu}); \boldsymbol{\mu}).}$$

Direct evaluation requires

$$\mathbf{u}_n \xrightarrow{\mathbf{V}} \underbrace{\mathbf{V} \mathbf{u}_n}_{\mathbb{R}^N} \xrightarrow{\mathbf{g}} \underbrace{\mathbf{g}(\mathbf{V} \mathbf{u}_n; \boldsymbol{\mu})}_{\mathbb{R}^N} \xrightarrow{\mathbf{V}^T} \mathbb{R}^n.$$

Hence the nonlinear evaluation still passes through the high-dimensional space $\mathbb{R}^N \rightarrow$ This is the lifting bottleneck. 33 / 35

Intro: Intrusive versus non-intrusive model reduction

Projection-based reduction requires more than snapshots.

To construct

$$\mathbf{A}_n(\boldsymbol{\mu}) = \mathbf{V}^T \mathbf{A}(\boldsymbol{\mu}) \mathbf{V}, \quad \mathbf{F}_n(\boldsymbol{\mu}) = \mathbf{V}^T \mathbf{F}(\boldsymbol{\mu}),$$

we need access to the full-model operators

$$\mathbf{A}(\boldsymbol{\mu}), \quad \mathbf{F}(\boldsymbol{\mu}),$$

or to routines that allow us to evaluate/project them $\rightarrow$ intrusive model reduction.

But in many applications the full model is available only as a black-box simulator

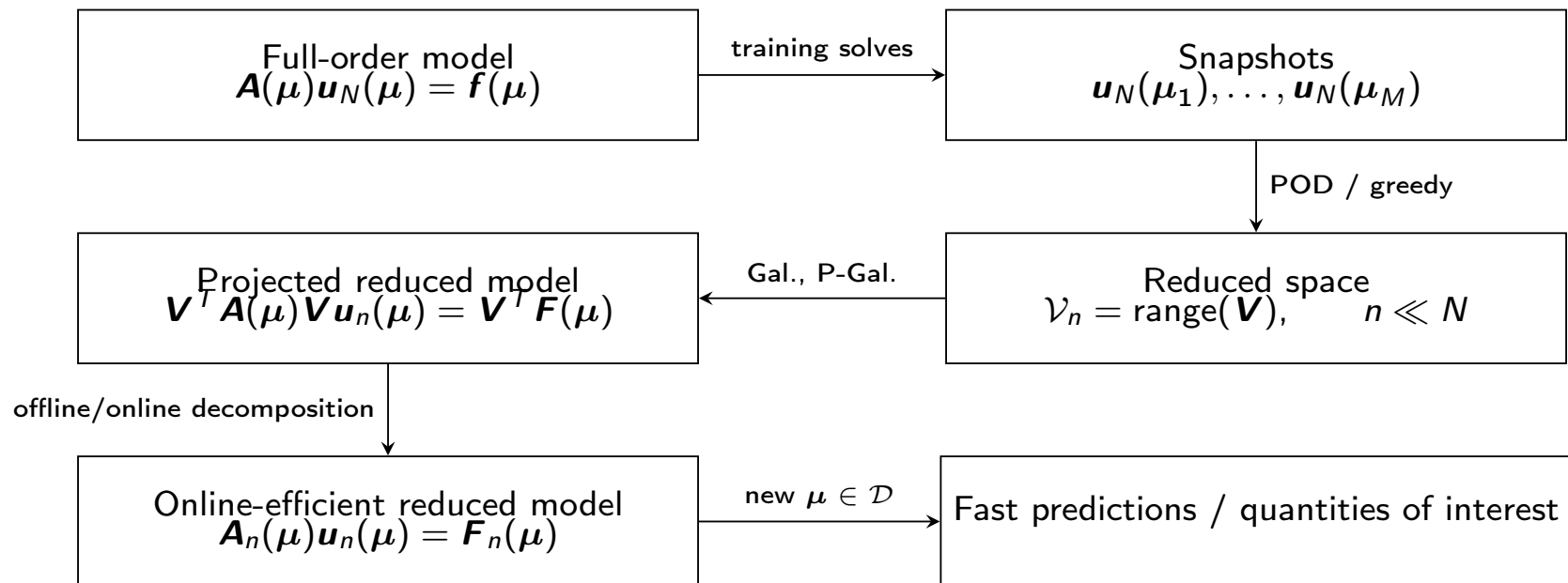
$$\boldsymbol{\mu} \longmapsto \mathbf{u}_N(\boldsymbol{\mu}).$$

We may be able to collect snapshots but have no access to

$$\mathbf{A}(\boldsymbol{\mu}), \quad \mathbf{f}(\boldsymbol{\mu}), \quad \text{Jacobians,} \quad \text{assembly routines.}$$

Then we cannot simply project the full-model operators via direct computations.

Intro: Outlook



Kolmogorov barrier



nonlinear approximations

Certification



**a posteriori error
estimation**

Online efficiency



empirical interpolation

Operator access



**non-intrusive MOR and
operator learning**

Formulation of full model

→ Where does $A(\mu)$, $u_N(\mu)$, $F(\mu)$ come from?

Abstract formulation:

↳ Consider a (real Hilbert) space V ; denote the inner product $(\cdot, \cdot)_V$ and the norm as $\|\cdot\|_V$

↳ Given a (parametrized) bilinear form

$$a: V \times V \times D \rightarrow \mathbb{R}$$

and a linear form

$$f: V \times D \rightarrow \mathbb{R}$$

and a given $\mu \in D$, seek $u(\mu) \in V$ such that

$$a(u(\mu), v; \mu) = f(v; \mu) \quad \forall v \in V.$$

↳ Additionally, often interested in a scalar quantity of interest (QoI)

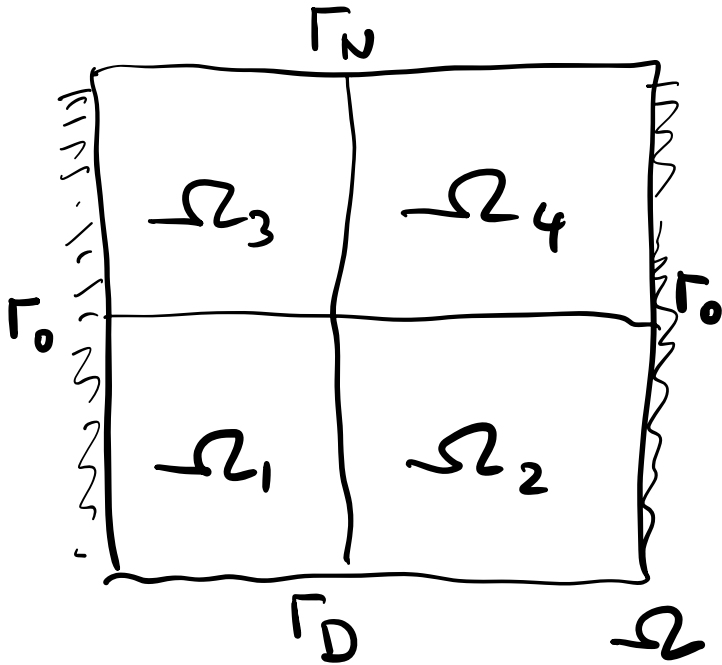
$$s(\mu) = \ell(u(\mu); \mu)$$

via linear output functional

$$\ell: V \times D \rightarrow \mathbb{R}.$$

Example: Thermal block

→ temperature in a "block"
with different material properties



• blocks $\Omega_i, i=1, \dots, p$

• parameter

$$\mu = [\mu_1, \dots, \mu_p]$$

$$\in D = [\mu_{\min}, \mu_{\max}]^p$$

• μ_i is heat conductivity coefficient of material in Ω_i

Conductivity is defined over Ω

$$k(x; \mu) = \sum_{i=1}^p \mu_i \chi_{\Omega_i}(x)$$

where χ_{Ω_i} denotes the char. function of Ω_i :

$$\chi_{\Omega_i}(x) = \begin{cases} 1, & x \in \Omega_i \\ 0, & x \notin \Omega_i \end{cases}$$

Boundary conditions

→ isolation on Γ_0

→ heat flux on Γ_N

→ impose temperature 0 of Γ_D

} → compute temperature
 $u(\mu): \Omega \rightarrow \mathbb{R}$
in block

↳ formulate as PDE

$$-\operatorname{div}(k(x;\mu) \nabla u(x;\mu)) = 0, \quad x \in \Omega$$

$$u(x;\mu) = 0, \quad x \in \Gamma_D$$

$$k(x;\mu) \nabla u(x;\mu) \cdot n(x) = 1, \quad x \in \Gamma_N$$

$$k(x;\mu) \nabla u(x;\mu) \cdot n(x) = 0, \quad x \in \Gamma_0$$

Weak form

↳ choose space V so that $(\hookrightarrow H_0^1)$

$$\forall v \in V: v|_{\Gamma_D} = 0$$

↳ Test equation with $v \in V$

$$\int -\operatorname{div}(k \nabla u) \cdot v \, dx = \int 0 \cdot v \, dx$$

⋮
Ship
⋮

$$\int k(x) \nabla u(x) \cdot \nabla v(x) \, dx = \int_{\Gamma_N} 1 \cdot v(x) \, dx$$

$$\begin{cases} \text{Bilinear form: } a(u, v; \mu) = \int k(x; \mu) \nabla u(x) \cdot \nabla v(x) \, dx \\ \text{Linear form: } f(v; \mu) = \int_{\Gamma_N} v(x) \, dx \end{cases}$$

Continuity and coercivity:

Def: For a fixed $\mu \in D$, the bilinear form a is continuous on $V \times V$ if there exists $f(\mu) < \infty$ such that

$$|a(w, v, \mu)| \leq f(\mu) \|w\|_V \|v\|_V$$

Denote the smallest possible constant

$$f(\mu) = \sup_{w \in V \setminus \{0\}} \sup_{v \in V \setminus \{0\}} \frac{|a(w, v, \mu)|}{\|w\|_V \|v\|_V}$$